

Lab #5 IWS: Physiology of the in situ Amphibian Heart

Individual assignment. *Text must be your own, but you may work together with your group to produce figures/tables. NOTE: This is a short format as compared to a full lab report.*

Instructions: You may download and edit the .docx version. In a Worksheet Assignment, we want to you zoom right in to the critical **intellectual content** of a full lab report without writing a lot. Think of it as an extended outline. Include: (1) Statements of Purpose, (2) Concept Check, (3) Mini Results, and (4) and Brief Discussion. If you feel uncomfortable leaving out the methods, add a brief Methods section before Results (optional).

(1) Write an opening sentence that is broad enough to capture all of the questions explored in this lab. (Often easier to write last). For example, for the Human ECG lab, the sentence might have been something like *"Heart function is critical to human performance and can be monitored using the echocardiogram (ECG)."*

(2) Identify and define the important ideas needed for the reader to understand the Hypotheses/ Statements of Purpose below (do this after completing #3). For example, in the Human ECG lab, the important ideas may have been: the cardiac cycle, systole and diastole, the components of the ECG in relation to the cardiac cycle. Write a sentence of introduction/definition for each important idea.

(3) Complete the statements of purpose to frame your thinking about this lab. *Study the examples below (make sure you include them), and complete the ones with blanks in a similar style and information content identifying parameters involved in the physiological mechanisms:*

Response to temperature _____

Response to stretch -- *We demonstrate the action of Starlings Law on the toad heart which predicts that the energy of cardiac contractions are proportional to the initial length of the cardiac muscle fibers.*

Responses to neurotransmitters and drugs -- *Although the toad heart has a myogenic pacemaker, cardiac performance can be modulated by the Central Nervous System via hormones and other chemical messengers. We demonstrate the action of potential excitatory and inhibitory messengers on cardiac performance.*

Responses to ion concentrations -- *Cardiac muscle contraction should be initiated by action potentials resulting from depolarization of cardiac muscle cells. By changing the _____ we should be able to demonstrate _____.*

Refractory period of cardiac muscle _____

Heart block _____

(4) Very Brief Results: Include either a figure or a table for each important result that addresses the hypotheses. **Write a sentence pointing out** what trends are shown in each display item (you may put these together into a paragraph). Follow the lab report guidelines: Don't forget figure captions, which are different than the paragraph text, and NO RAW DATA.

(5) Brief Discussion: In paragraph form, **briefly discuss** the **main take-aways** that you learned from these experiments. Start by summarizing the main results. Then discuss. Use specific results that back up your statements or speculate on the significance of the results. Organize by hypotheses above. You can see the guidelines for the full lab report if you wish.